

Topic: Stack using Singly Linked List

Q. Explain how a Stack can be implemented using a Singly Linked List. Provide the logic for the Push and Pop operations.

****Definition to Remember****

A Stack
dynamic
stack.

Stack Implementation using Singly Linked List

To maintain $O(1)$ time complexity for both Push and Pop operations, all insertions and deletions must occur at the beginning (head) of the linked list.

| Stack Operation | Linked List Equivalent |

| :--- | :-

2. Push Operation (Insert at Beginning)

Adds a new item to the top of the stack.

Algorithm:

text

3. Pop Operation (Delete from Beginning)

Removes and returns the item from the top of the stack.

Algorithm:

text

4. Advantages of Linked List Implementation

* ****Dynamic Size:**** The stack can grow and shrink dynamically. It is limited only by system memory, unlike an array which has a fixed pre-defined size.

* ****No**

*****Must-Write Points (for 10 marks)*****

1. Stack

*****Quick Recall*****

`Stack v
(O(1))`